Nested GLS Profile Spectrum: First Smoke Test

June 21, 2026

Purpose

This note records what is needed to compute the full regular eigenspectrum for nested tests in the profile-Hessian paradigm, and a first finite-sample smoke test in a continuous GLS setting. The companion scripts are

`relative-and-nested-fit-check.py` and `gls_nested_profile_mc.py`.

The finite-sample script uses the C++ helper `weighted_chisq_quantile_cli.cpp`, which calls `magmaan`'s `robust::weighted_chisq_quantile` and `robust::weighted_chisq_upper` for deterministic mixture quantiles. The only Monte Carlo in the smoke is the finite-sample data generation.

1 Required Objects

Let x be the first-stage vector and let

$$\sqrt{n}(\hat{x} - x_0) \Rightarrow Z, \qquad Z \sim N(0, \Gamma_x).$$

For each model M, define the profiled value

$$v_M(x) = \min_{\theta} f_M(\theta, x).$$

The nested contrast is

$$c(x) = v_B(x) - v_A(x),$$

where B is the restricted model and A is the larger model. Under the regular nested pseudo-null,

$$c(x_0) = 0, \qquad \nabla c(x_0) = 0.$$

The reference law is therefore

$$n\,c(\hat{x}) \Rightarrow \frac{1}{2}Z'(Q_B - Q_A)Z$$

if f is scaled as a half-discrepancy, or equivalently

$$2n\,c(\hat{x}) \Rightarrow Z'(Q_B - Q_A)Z.$$

Thus the full spectrum is obtained from the self-adjoint version of

$$(Q_B - Q_A)\Gamma_x.$$

In implementation terms, the steps are:

1. Build a common first-stage vector x and covariance Γ_x.
2. Compute the profiled score $s_M(x) = \nabla_x v_M(x)$ for each model.
3. Differentiate it to get $Q_M = D_x s_M(x_0)$.
4. Verify the pseudo-null conditions $v_B = v_A$ and $s_B = s_A$.
5. Eigensolve $\Gamma_x^{1/2}(Q_B - Q_A)\Gamma_x^{1/2}$.

The old $U_B - U_A$ spectrum is the special exact-fit/fixed-weight/Gauss–Newton case. It is not the general object.

2 GLS Smoke Design

The smoke uses the linear covariance example from `relative-and-nested-fit.tex`. There are four observed variables.

$$B: \text{ compound symmetry: one variance and one covariance,}$$

$$A: \text{ one common variance and free covariances.}$$

The test is equality of the covariances. The normal-theory GLS weight is estimated from the first-stage covariance:

$$W(u) = \Gamma_{\mathrm{NT}}(u)^{-1}.$$

To make the nested pseudo-null exact while keeping both models misspecified, the population covariance is diagonal with unequal variances:

$$\Sigma_0 = I + \epsilon\,\mathrm{diag}(d), \qquad d = (1, -1, 1, -1).$$

Both models share the same pseudo-true zero covariance structure, so

$$v_B(x_0) - v_A(x_0) = 0, \qquad s_B(x_0) - s_A(x_0) = 0,$$

but the common-variance part is misspecified when $\epsilon \neq 0$.

3 Population Spectrum

For $\epsilon = 0.15$, the population script reports:

$$v_B - v_A = 0, \qquad \|s_B - s_A\| = 0,$$

and the profile/value finite-difference Hessians agree to 1.17×10^{-7}. The classical fixed-weight spectrum is

$$(1, 1, 1, 1, 1),$$

whereas the profile-Hessian spectrum is

$$(1.4082,\ 0.9139,\ 0.9139,\ 0.9139,\ 0.5057), \qquad \mathrm{tr} = 4.6557.$$

So even in this linear model with no model-curvature term, estimating the GLS weight changes the reference law under fixed misspecification.

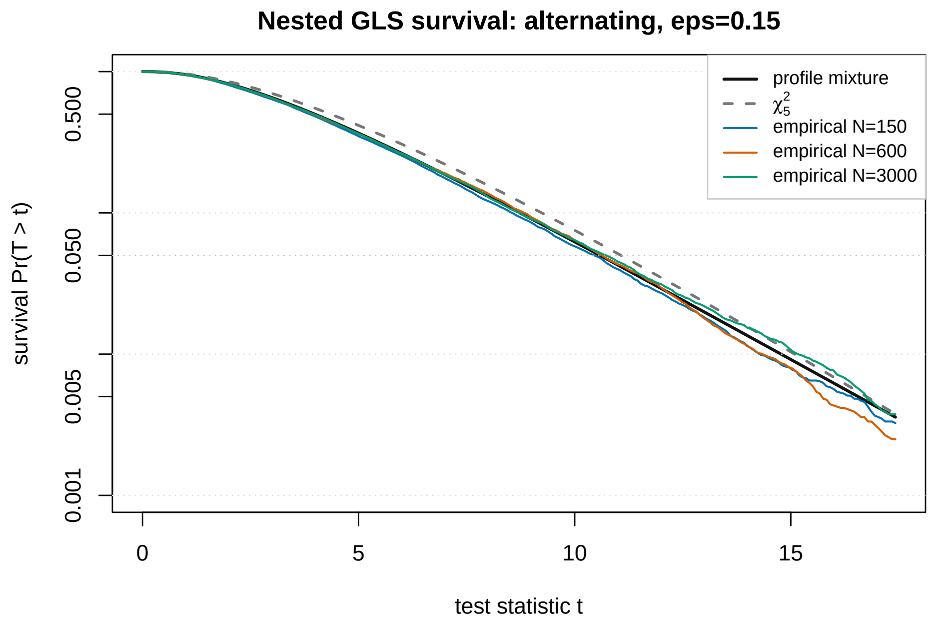
4 Finite-Sample Smoke

The Monte Carlo uses 6000 replications for each sample size, simulating normal data from the same Σ_0 and refitting both GLS profiles with $W(\hat{u})$. The reference quantiles are computed once from the population spectrum.

Reference	q50	q90	q95	q99
Profile spectrum	3.9817	8.7316	10.5854	14.7734
Classical U / χ^2_5	4.3515	9.2364	11.0705	15.0863

N	empirical q50	empirical q90	empirical q95	tail at profile q95	tail at χ^2_5 q95
150	3.8651	8.4670	10.1870	0.0437	0.0352
300	3.9323	8.4929	10.3074	0.0457	0.0385
600	3.9239	8.6244	10.4148	0.0475	0.0397
1200	3.9793	8.7283	10.5150	0.0487	0.0405
3000	3.8889	8.5904	10.5366	0.0487	0.0395

The profile cutoff is close to nominal once N is moderate. The ordinary χ^2_5 cutoff remains conservative here because the profile spectrum has a smaller upper tail cutoff than the exact-fit spectrum.



The survival-curve smoke uses 12000 replications for $N = 150, 600, 3000$ and compares the whole tail curve against the two deterministic references. The maximum absolute survival gap over the plotted grid was

N	150	600	3000
profile mixture	0.0201	0.0085	0.0065
χ^2_5	0.0696	0.0593	0.0567

for this seed and grid. This is a better diagnostic than only checking the 0.95 cutoff, because it shows the ordinary chi-square reference is not merely off at one threshold; it is visibly too far to the right across the upper tail.

5 A High-Contrast Stress Test

The population scan in `gls_nested_profile_scan.py` searches diagonal pseudo-null misspecifications before running Monte Carlo. The useful heuristic is simple: the fixed-weight U spectrum stays close to the exact-fit reference, whereas the profile spectrum changes through the derivative of the estimated GLS weight. Those derivative terms are driven by the pseudo-true residual, so large variance heterogeneity is the first place to look.

One strong $p = 4$ stress case uses a linear diagonal pattern with $\epsilon = 0.85$, so the marginal variances range from 0.15 to 1.85. The data-generating law is

$$Y \sim N_4(0, \Sigma_0), \qquad \Sigma_0 = \begin{pmatrix} 0.1500 & 0 & 0 & 0 \\ 0 & 0.7167 & 0 & 0 \\ 0 & 0 & 1.2833 & 0 \\ 0 & 0 & 0 & 1.8500 \end{pmatrix}.$$

The larger model A is common variance with free covariances,

$$\Sigma_A(a, c) = \begin{pmatrix} a & c_{12} & c_{13} & c_{14} \\ c_{12} & a & c_{23} & c_{24} \\ c_{13} & c_{23} & a & c_{34} \\ c_{14} & c_{24} & c_{34} & a \end{pmatrix},$$

and the restricted model B is compound symmetry,

$$\Sigma_B(a, b) = \begin{pmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{pmatrix}.$$

At the population point the normal-theory GLS weight is $W_0 = \Gamma_{\mathrm{NT}}(\Sigma_0)^{-1}$. Because Σ_0 is diagonal, the off-diagonal moment weights are diagonal and all true covariances are zero, so both models choose zero covariance:

$$c_{ij}^* = 0, \qquad b^* = 0.$$

The common variance is the GLS projection of the four true variances onto one number,

$$a^* = \frac{\sum_i w_i \sigma_i^2}{\sum_i w_i}, \qquad w_i = \frac{1}{2\sigma_i^4},$$

which gives

$$a_A^* = a_B^* = 0.1983847695.$$

Thus

$$\Sigma_A^* = \Sigma_B^* = 0.1983847695\, I_4,$$

while both models are badly misspecified as approximations to Σ_0. The population half-discrepancies are equal,

$$f_A(\Sigma_0) = f_B(\Sigma_0) = 0.5346995918,$$

so this is a genuine nested pseudo-null: the less restricted and more restricted models share the same pseudo-true covariance even though neither model is true.

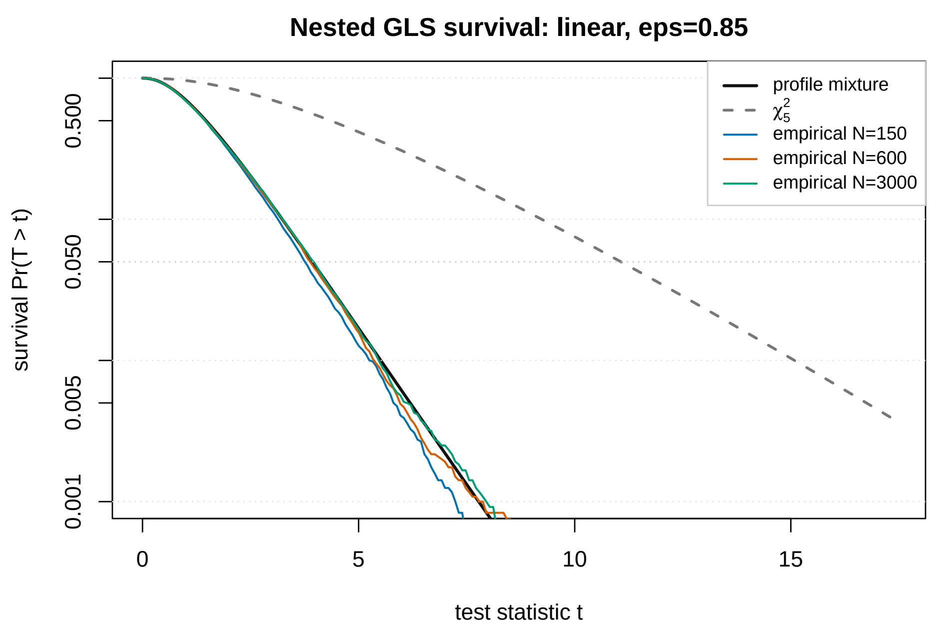
The profile spectrum is

$$(0.5401,\ 0.3781,\ 0.3279,\ 0.2767,\ 0.1990),$$

giving a profile 0.95 cutoff of 3.9305, versus the ordinary χ^2_5 cutoff 11.0705. The finite-sample smoke again used 12000 replications:

N	150	600	3000
$\Pr(T > q_{0.95}^{\text{profile}})$	0.0404	0.0475	0.0508
$\Pr(T > q_{0.95}^{\chi^2_5})$	0.0000	0.0001	0.0000

So the ordinary chi-square reference is not mildly conservative in this stress case; it is effectively unusable for a nominal 0.05 test.



A round three-indicator conservative example

The phenomenon is not tied to four indicators, and it does not require non-round decimals. For $p = 3$, take the diagonal pseudo-null

$$\Sigma_0 = \mathrm{diag}(4,\ 4,\ 1).$$

The same model pair is used: model A has one common variance and three free covariances, while model B has one common variance and one common covariance. The population GLS projections again coincide,

$$\Sigma_A^* = \Sigma_B^* = \frac{4}{3} I_3, \qquad f_A(\Sigma_0) = f_B(\Sigma_0) = 0.25.$$

The profile spectrum has only two positive weights,

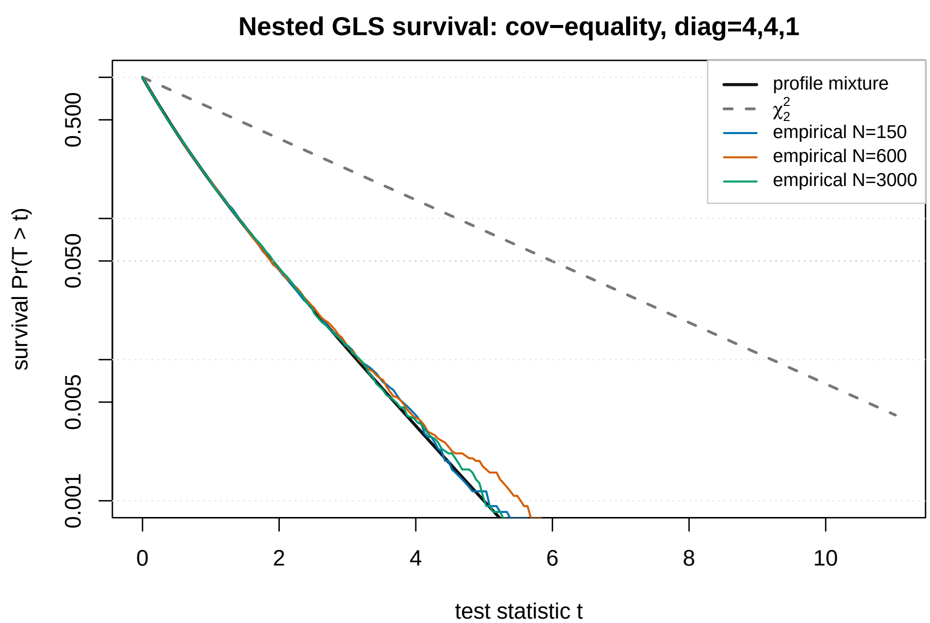
$$(0.4444,\ 0.1481),$$

so the profile 0.95 cutoff is 1.9034, while the ordinary χ_2^2 cutoff is 5.9915. Under the profile law, the ordinary cutoff has tail probability 0.00030. In 12000 replications, the profile cutoff rejected at

$$0.0478, \quad 0.0480, \quad 0.0508$$

for $N = 150, 600, 3000$, while the ordinary chi-square cutoff rejected at

$$0.0007, \quad 0.0003, \quad 0.0003.$$



A round one-degree liberal example

The same mechanism can also move in the opposite direction; the conservative examples above are not a theorem about GLS. Keep $p = 3$, but use the simpler nested contrast that tests one covariance:

$$B:\ \Sigma_B(a) = aI_3, \qquad A:\ \Sigma_A(a,c) = \begin{pmatrix} a & c & 0 \\ c & a & 0 \\ 0 & 0 & a \end{pmatrix}.$$

Now take

$$\Sigma_0 = \operatorname{diag}(4,\ 4,\ 7).$$

The true covariance between the first two variables is zero, and the first two variances are equal. The extra covariance parameter in A therefore remains zero at the population optimum even though both models are misspecified by the third variance. The GLS projection is

$$\Sigma_A^* = \Sigma_B^* = \frac{84}{19} I_3, \qquad f_A(\Sigma_0) = f_B(\Sigma_0) = 0.0394736842.$$

This is again a genuine nested pseudo-null: the value and profile gradient differences are zero at Σ_0. The profile spectrum is

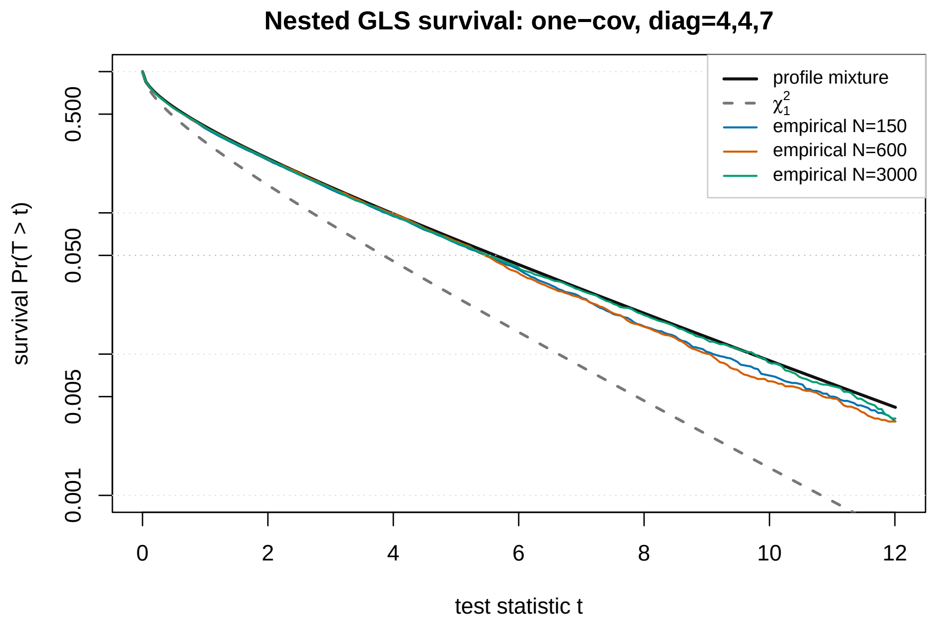
$$(1.4654),$$

so the profile 0.95 cutoff is 5.6292, while the ordinary χ_1^2 cutoff is 3.8415. Under the profile law, the ordinary cutoff has tail probability 0.1054, so the nominal 0.05 test is asymptotically about a 10.5 percent test. In 12000 replications, the profile cutoff rejected at

$$0.0466, \quad 0.0453, \quad 0.0478$$

for $N = 150, 600, 3000$, while the ordinary chi-square cutoff rejected at

$$0.1009, \quad 0.1048, \quad 0.1031.$$



Growth with the number of indicators

The conservative failure is not a finite-p artifact. A clean construction is block replication. Take K independent copies of the round conservative three-indicator pseudo-null block above, and test the same within-block covariance equality restriction in each block. This is a block-diagonal SEM sequence rather than the single global compound-symmetry family, but it is enough to show that the issue does not disappear with more indicators. The ordinary reference has $2K$ degrees of freedom. The profile reference is

$$\sum_{k=1}^{K}\{0.4444\,\chi^2_{1k,1} + 0.1481\,\chi^2_{1k,2}\}.$$

Its mean is $K(0.4444 + 0.1481) = 0.5926K$, whereas the ordinary χ^2_{2K} reference has mean $2K$. Therefore the ordinary 0.95 cutoff, which is $2K+O(\sqrt{K})$, lies far to the right of the true profile mixture mean by an order-K amount. By concentration,

$$\Pr_{\text{profile}}\{T > q_{0.95}(\chi^2_{2K})\} \to 0.$$

So the nominal 0.05 chi-square test can be made arbitrarily conservative, in the sense that its true size tends to zero as the number of replicated indicator blocks grows. The absolute size error is then as large as possible for a conservative 0.05-level test.

This is intentionally an extreme design. The practical next question is how large the gap remains under less ill-conditioned but still realistic covariance patterns, and then under ordinal DWLS where the first-stage weight inputs are more variable.

6 Magmaan Machinery

The reusable library pieces are now:

$$\texttt{robust::compute\_profile\_contrast\_spectrum}(Q, \Gamma),$$

which returns the positive eigenvalues of $\Gamma^{1/2} Q \Gamma^{1/2}$ and flags genuinely negative directions, and

$$\texttt{robust::weighted\_chisq\_upper}, \qquad \texttt{robust::weighted\_chisq\_quantile},$$

which provide the central positive weighted-χ^2 reference law and its cutoffs. The old `robust::imhof_upper` symbol remains as a compatibility alias, but the new call sites use the distribution-level name.

7 What This Suggests

For GLS, the full nested eigenspectrum is not hard in principle. The hard part is not the eigensolve; it is computing Q_M reliably. Finite differences of the profile score are enough for research scripts. The next production step is to expose analytic or automatic profile derivatives for:

$$s_M(x), \qquad Q_M = D_x s_M(x).$$

For categorical DWLS, the same steps apply, but x must include the weight inputs such as $\gamma = \mathrm{diag}(\Gamma_u)$. That makes Q_M a block Hessian over (u, γ) rather than only over ordinal moments. The GLS smoke is therefore the right first implementation target: it validates the profile machinery before the ordinal first-stage stack is made larger.